



TEST CODE **01212020**

FORM TP 2012005

JANUARY 2012

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

CHEMISTRY

Paper 02 – General Proficiency

2 hours and 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX compulsory questions in TWO sections.
2. Write your answer to EACH question in the space provided in this answer booklet.
3. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
4. Return this booklet at the end of the examination.
5. The use of silent, non-programmable calculators is allowed.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer ALL questions in this section.

Write your answers in the spaces provided in this booklet.

Do NOT spend more than 30 minutes on Question 1.

1. A group of students conducted three experiments: Experiment 1, Experiment 2 and Experiment 3.

- (a) **Experiment 1** was carried out to determine the heat of neutralization for the reaction between sodium hydroxide and sulphuric acid.

The procedure is given below and the data collected are given in Table 1.

Procedure for Experiment 1

- 25 cm³ of dilute sodium hydroxide was measured and poured into a styrofoam cup.
- 25 cm³ of sulphuric acid was poured into a burette.
- The solutions were left for a while until they reached room temperature and their temperatures recorded.
- The sulphuric acid was added to the sodium hydroxide solution 3 cm³ at a time.
- The mixture was stirred on each addition of acid and the maximum temperature, in °C, was recorded in Table 1.

TABLE 1: RESULTS FOR EXPERIMENT 1

Volume of Sulphuric Acid Added (cm ³)	0	3	6	9	12	15	18	21
Temperature of Mixture (°C)	25	30	36	41	43	38	31	22

- (i) Define the term 'heat of neutralization'.

(2 marks)

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- (ii) Use the data in Table 1 to plot a graph of temperature against volume of acid for Experiment 1 using the axes provided on page 5. The first two points have been plotted for you. **(3 marks)**
- (iii) From the graph, determine the volume of sulphuric acid required to completely neutralize 25 cm³ of sodium hydroxide. **(1 mark)**
- (iv) Given that the difference in temperature from the start of the reaction to the point of neutralization is 18 °C, calculate the heat change at the point of neutralization for the reaction between sodium hydroxide and sulphuric acid.

[The specific heat capacity of the solution is 4.2 kJ kg⁻¹ K⁻¹. Assume that the density of the solution is 1 g cm⁻³, $\Delta H = m \times c \times \Delta T$.]

(3 marks)

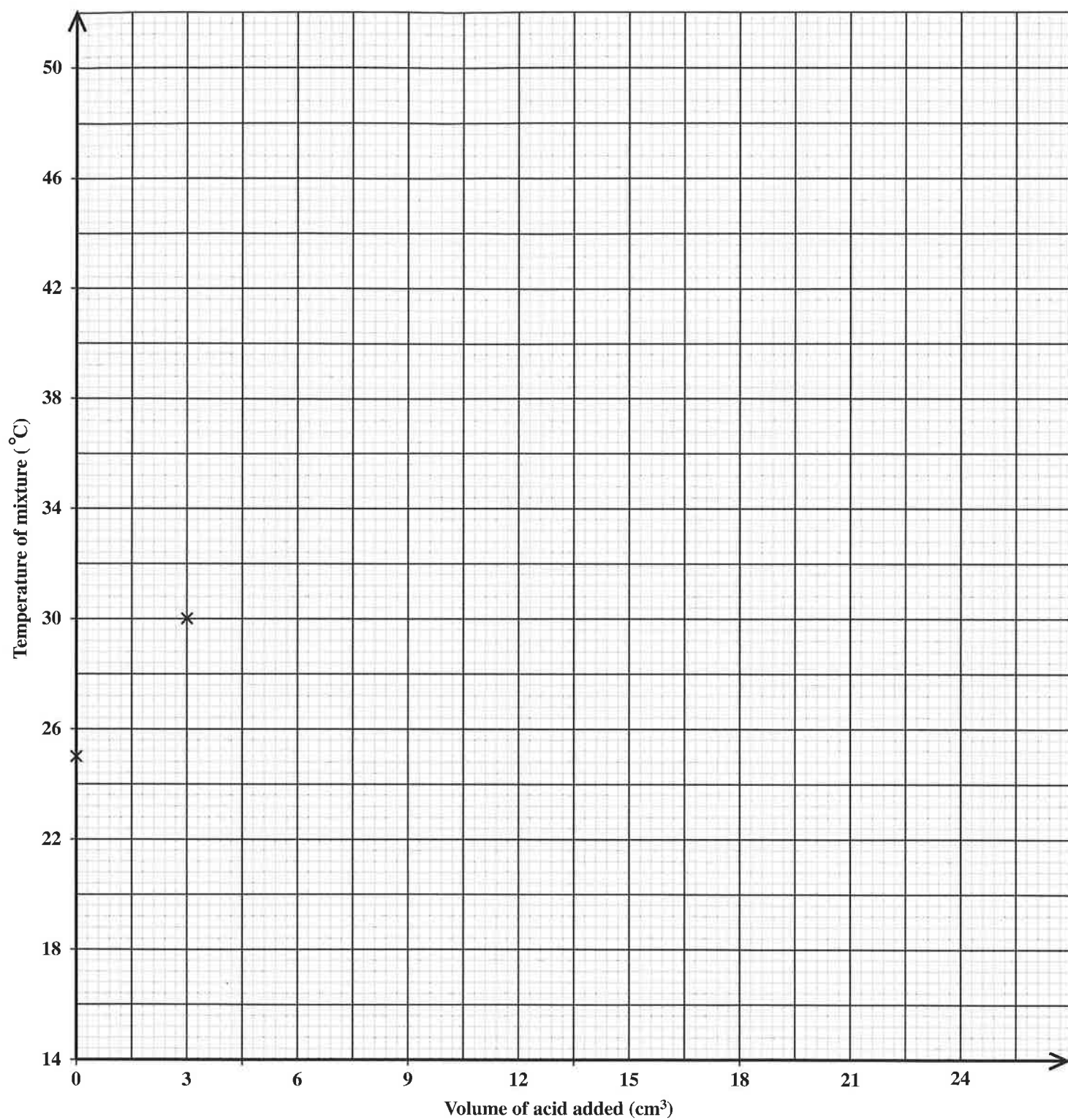


Figure 1. Temperature of mixture versus volume of acid

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- (b) **Experiment 2** was carried out to determine the heat of solution made by mixing 8 g of potassium nitrate in 50 cm³ of water.

(i) Define the term 'heat of solution'.

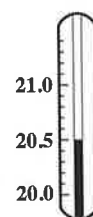
(2 marks)

- (ii) Figure 2 shows the thermometer readings for the initial and final temperatures obtained for Experiment 2. Record the reading, in °C, on each thermometer in the space provided.

Initial Temperature



Final Temperature



Initial Temperature _____

Final Temperature _____

Figure 2. Thermometer readings for Experiment 2

(1 mark)

- (iii) Write a suitable procedure that could be carried out for Experiment 2.

(3 marks)

- (iv) Draw a labelled energy profile diagram for the reaction taking place in Experiment 2.

(3 marks)

- (c) **Experiment 3** involved reactions between a colourless unknown Solution X and three aqueous solutions, iron(II) nitrate, acidified potassium manganate(VII) and potassium iodide. Two of the three observations made are recorded in Table 2.

TABLE 2: REACTIONS OF SOLUTION X

Reactions of Solution X		
Aqueous Iron(II) Nitrate	Acidified Potassium Manganate(VII)	Aqueous Potassium Iodide
The solution changes from pale green to pale yellow.	The solution changes from purple to colourless.	Observation 3

Use the data in Table 2 to answer the following questions.

- (i) Write a **balanced** half-equation to represent the change which takes place when Solution X reacts with aqueous iron(II) nitrate.

(2 marks)

- (ii) Explain whether Solution X is behaving as a reducing agent, an oxidizing agent or both.

(3 marks)

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- (iii) Describe Observation 3 as indicated in Table 2.

(2 marks)

Total 25 marks

2. An experiment was carried out at 25 °C by dissolving an unknown mass of limestone granules into excess 1 mol dm⁻³ hydrochloric acid. The volume of gas produced over a period of 6 minutes was measured using a gas syringe. The result was plotted as shown in Figure 3.

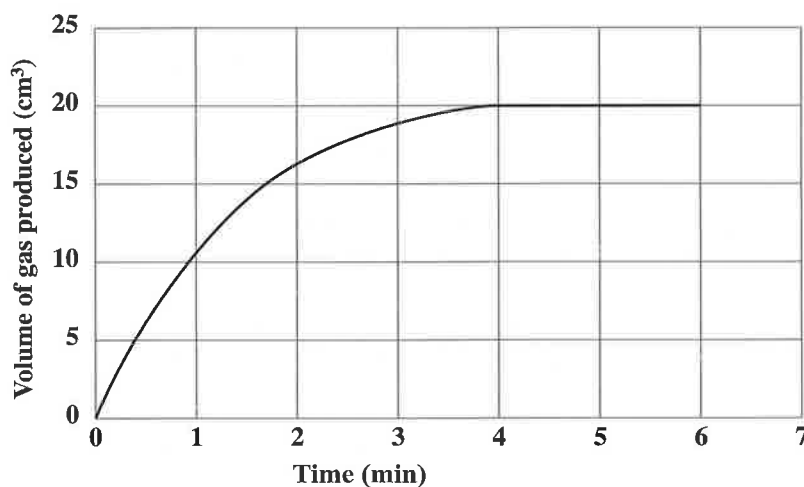


Figure 3. Volume of gas produced against time

- (a) Assuming that the limestone granules are pure calcium carbonate, write a **balanced** equation for its reaction with hydrochloric acid.

(2 marks)

- (b) Use the data from Figure 3
- (i) to determine the **total** volume of gas produced

(1 mark)

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- (ii) to calculate the moles of gas produced at STP
[1 mole of any gas at STP has a volume of 22 400 cm³.]

(1 mark)

- (iii) to calculate the mass of limestone granules used.
[Relative molecular mass of CaCO₃ is 100.]

(2 marks)

- (c) (i) What is meant by the term 'rate of reaction'?

(1 mark)

- (ii) Besides temperature, identify TWO **other** factors that can affect the rate of the reaction.

(2 marks)

- (iii) a) If the temperature at which the reaction was carried out was changed from 25 °C to 40 °C, would the rate of reaction increase or decrease?

_____ (1 mark)

- b) Explain your answer.

_____ (2 marks)

- (d) Like sodium chloride, calcium carbonate has a giant ionic crystalline structure.

- (i) Is it LIKELY that calcium carbonate will conduct electricity in its solid state?

_____ (1 mark)

- (ii) Give a reason for your answer.

_____ (2 marks)

Total 15 marks

3. (a) Hydrocarbons are compounds that contain the elements, hydrogen and carbon. Name TWO natural sources of hydrocarbons.

(2 marks)

- (b) Fractional distillation of crude oil produces several fractions, containing a range of compounds with varying numbers of carbon atoms. Three of these fractions are shown in Table 3.

TABLE 3: THREE FRACTIONS FROM CRUDE OIL

Fraction	1	2	3
Number of Carbon Atoms	C1 – C4	C12 – C18	C20 – C40

- (i) State the name of Fraction 1 **OR** Fraction 3.

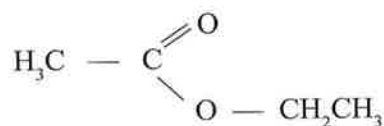
Fraction number: _____

Fraction name: _____
(1 mark)

- (ii) State ONE use of Fraction 2.

(1 mark)

- (c) Compound X is sweet smelling, and is hydrolysed by dilute hydrochloric acid.



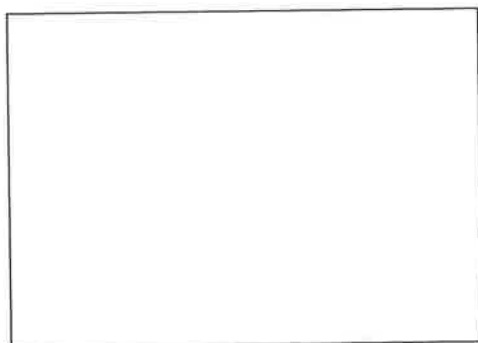
Compound X

- (i) State the name of Compound X and the homologous series to which it belongs.

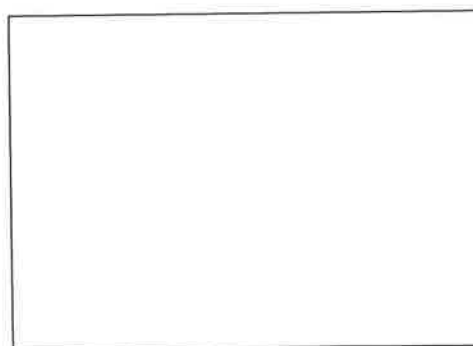
Name of Compound X: _____ (1 mark)

Homologous series: _____ (1 mark)

- (ii) Write the FULLY DISPLAYED structural formulae of the two molecules that are produced when Compound X is hydrolysed.



Structure 1



Structure 2

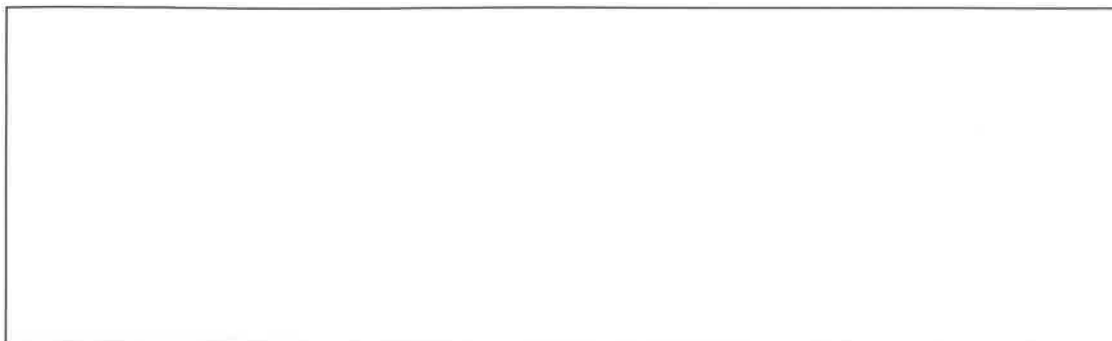
(2 marks)

(d) (i) Glucose, a monosaccharide, can be represented by the HO——OH unit.

a) Name a polymer of glucose.

Name of polymer: _____ (1 mark)

b) Using three glucose units, draw the partial structure of the polymer in the space provided.



Partial structure of polymer of glucose

(2 marks)

(ii) A student is asked to distinguish between a sample of glucose and a sample of its polymer. The student adds a portion of glucose and a portion of the polymer to two separate beakers of water, and stirs the contents of the beakers.

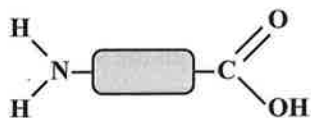
State the expected observations in EACH case.

Glucose: _____

Polymer: _____

(2 marks)

(e) The monomer shown below undergoes condensation polymerisation. Name the type of polymer formed and give ONE use for this type of polymer.



Monomer

Type of polymer: _____ (1 mark)

Use: _____ (1 mark)

Total 15 marks

SECTION B

Answer ALL questions in this section.

Write your answers in the space provided after EACH question in this answer booklet.

4. Selected elements in the Periodic Table are presented in Figure 4.

I	II		III	IV	V	VI	VII	VIII
Li	Be			C	N	O	F	
Na	Y				P	S	Cl	
K							Br	
X							I	

Figure 4. Selected elements of the Periodic Table

- (a) (i) Write the electronic configuration of sodium and state a reason for its placement in Period 3. (2 marks)
- (ii) Sodium and potassium react vigorously with water at room temperature.
- a) Write a **balanced** equation for the reaction of sodium with water. (2 marks)
- b) Outline ONE test that can be used to identify any gas that is produced during the reaction. (2 marks)
- c) An unknown element X is shown in Figure 4. Predict the reactivity of X with water as compared with sodium and potassium. (1 mark)
- (iii) Element Y and sulphur react with oxygen gas to form oxides. Both of these oxides are soluble in water to form colourless solutions. Give ONE test that can be used to distinguish the solution of the oxide of Element Y from the solution of the oxide of sulphur. (2 marks)
- (b) (i) During an experiment, chlorine gas, $\text{Cl}_2(\text{g})$, is bubbled into a green solution of acidified iron(II) sulphate. The solution changes gradually from green to yellow. Write a **balanced** IONIC equation for the overall reaction, and identify the species being oxidised and reduced. (4 marks)
- (ii) In a second experiment, iodine is stirred with another solution of iron(II) sulphate. No observable change takes place. Account for the difference in reactivity of chlorine and iodine towards iron(II). (2 marks)

Total 15 marks

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Write your answer to Question 4 here.

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5. (a) Iron is extracted from its ore (Fe_2O_3) by thermal reduction using carbon monoxide.
- (i) Describe the process for obtaining iron from its ore. In your description, identify the materials used and include a chemical equation to show how iron is produced. **(6 marks)**
- (ii) Carbon and carbon monoxide are both able to reduce the oxide ore to produce iron. Suggest a suitable explanation why carbon monoxide is used to reduce the iron ore instead of carbon. **(1 mark)**
- (b) A chemist analysed a newly discovered oxide of a Metal M and found that it had the following properties:
- The oxide of Metal M was soluble in sodium hydroxide.
 - The formula for the oxide of Metal M was MO .
 - When the oxide (MO) was heated with carbon at high temperatures, no reaction occurred.
 - Metal M displaced aluminium from a solution of Al^{3+} ions.
- (i) Write suitable equations for the action of heat on the
- a) hydroxide of M
- b) nitrate of M. **(4 marks)**
- (ii) Suggest a suitable method for extracting Metal M from its ore and explain your answer. **(3 marks)**
- (iii) Suggest ONE chemical property of Metal M. **(1 mark)**

Total 15 marks

Write your answer to Question 5 here.

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- Dough has a mixture of flour, water and a leavening agent as the major ingredients. During the making of dough, a number of important physical and chemical changes take place.

- (a) Discuss the role of EACH of the following agents in the making of dough:

- (i) Water (2 marks)

- (ii) Leavening agent (baking powder and yeast) (2 marks)

- (iii) Heat (2 marks)

- (b) Write THREE relevant chemical equations to support your answer in (a). Include AT LEAST ONE equation from the action of yeast and ONE from the action of baking powder. (6 marks)

- (c) Flour usually consists of 55–80% carbohydrates (starches). Iodine can be used to test for the presence of starches.

There is no blue-black colouration when flour, which is stored in a hot, humid room for a long time, is tested with iodine. Suggest a possible explanation for this.

(3 marks)

Total 15 marks

Write your answer to Question 6 here.

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Write your answer to Question 6 here.

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.